



Addis Ababa Institute of Technology
Department of Software Engineering
Second Assignment of computer programming

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SECTION:- 2

1. Write a python program that accepts a positive number and prints the sum of the digits in the number. i.e. if the number is 576, it should print $18 = 5 + 7 + 6$.

☆ Write the code below

☆ Write the explanation below

Answer

That accepts a positive number and prints the sum of the digits in the number in Python:

☆ **Write the code below**

```
num = int(input("Enter a positive number: ")) # take input from user for a positive number
```

```
sum = 0
```

```
while num > 0: # loop until all digits have been added to sum
```

```
    digit = num % 10 # get the rightmost digit
```

```
    sum += digit # add the digit to the sum
```

```
    num //= 10 # remove the rightmost digit from the number
```

```
# print the sum of digits of the given number
```

```
print("The sum of digits of", num, "is", sum)
```

☆ **Write the explanation below**

- We first take input from the user for a positive number using the `input()` function and convert it to an integer using the `int()` function. This number is stored in the variable `num`.
- We initialize the variable `sum` to 0, which we will use to store the sum of digits of the given number.
- We use a `while` loop to loop through the digits of the given number. The loop will iterate until all the digits have been added to the `sum` variable.
- In each iteration of the loop, we get the rightmost digit of the given number by using the modulus operator `%` with a divisor of 10. This digit is then stored in the `digit` variable.
- We add the `digit` to the `sum` variable.- We remove the rightmost digit from the given number by using floor division operator `//` with a divisor of 10. This updated `num` value will be used in the next iteration of the loop.
- After the `while` loop finishes looping through all the digits, we print the sum of digits of the given number using the `print()` function and string concatenation to display the variables used in the message.

2. Write a python program that accepts a number and prints the sum of its factors.

☆ Write the code here

☆ Write the description

Answer

The code that accepts a number and prints the sum of its factors in Python:

☆ **Write the code here**

```
num = int(input("Enter a number: ")) # take input from user for the number

sum = 0

# loop through all numbers from 1 to the given number
for i in range(1, num+1):

    if num % i == 0: # check whether the current number is a factor of the given
number

        sum += i # add the factor to the sum variable

# print the sum of factors of the given number
print("The sum of factors of", num, "is", sum)
```

☆ **Write the description**

- We first take input from the user for the number using the `input()` function, and convert it to an integer using the `int()` function. This number is stored in the variable `num`.
- We initialize the variable `sum` to 0, which we will use to store the sum of factors of the given number.
- We then use a `for` loop to loop through all numbers from 1 to the given number (`num`). In each iteration of the loop, the variable `i` takes on the value of the current number being examined.
- We check whether the current number (`i`) is a factor of the given number (`num`) using the modulus operator `%`. If the remainder of the division operation

between `num` and `i` is equal to 0, then `i` is a factor of `num`, so we add it to the variable `sum`.

- After the `for` loop finishes looping through all numbers from 1 to `num`, we print the sum of factors of the given number using the `print()` function and string concatenation to display the variables used in the message.

3. Write a python program that generates a random number in the range 0 to 100 and prompts the user to guess and enter the number continuously until the user's input matches the randomly generated number. The program should inform the user if his/her guess is less/more than the random number. if the user can't guess the value by entering at most 7 guesses the program should print GAME OVER

☆ Write the code here

☆ Write the description below

Answer

The code that generates a random number in the range 0 to 100 and prompts the user to continuously guess the number until they get it right or reach a limit of 7 guesses in Python:

☆ **Write the code here**

```
import random
```

```
number = random.randint(0, 100) # generate a random number between 0 and 100
```

```
guesses = 0 # initialize the number of guesses to 0

while guesses < 7: # loop until the user runs out of guesses
    guess = int(input("Guess the number between 0 and 100: ")) # take input from
    user for the guess

    guesses += 1 # increment the number of guesses

    if guess == number: # check if the guess is correct
        print("Congratulations, you guessed the number in", guesses, "guesses!")
        break # exit the loop since the guess is correct

    elif guess < number:
        print("Your guess is less than the number, try again.")

    else:
        print("Your guess is more than the number, try again.")

if guesses == 7: # check if the user has run out of guesses
    print("GAME OVER. The number was", number)
'''
```

☆ **Write the description below**

- We first import the ``random`` module to generate random numbers.
- We generate a random number between 0 and 100 using the ``randint()`` function from the ``random`` module, and store it in the variable ``number``.
- We initialize the variable ``guesses`` to 0, which we will use to keep track of the number of guesses the user has made.
- We use a ``while`` loop to prompt the user to keep guessing until they either guess the correct number or reach a limit of 7 guesses.
- In each iteration of the loop, we take input from the user for their guess using the ``input()`` function and convert it to an integer using the ``int()`` function. This guess is stored in the variable ``guess``.
- We increment the ``guesses`` variable by 1 to keep track of the total number of guesses the user has made.
- We use a series of ``if`` statements to check whether the user's guess is correct or not. If the guess is correct, we print a congratulatory message to the user with the number of guesses it took them to guess the correct number, and we break the loop since the game is over. If the guess is not.

4. An Armstrong number of three digits is an integer such that the sum of the cubes of its digits is equal to the number itself. For example, 371 is an Armstrong number

Since,

$$3^3 + 7^3 + 1^3 = 371.$$

Write a program to find all Armstrong number in the range of 0 and 999

☆ Write the code here

☆ Write the description below

Answer

one way to implement the program to find all Armstrong numbers between 0 and 999 in Python:

☆ **Write the code here**

```
for num in range(0, 1000):  
    order = len(str(num))  
    sum = 0  
    temp = num  
    while temp > 0:  
        digit = temp % 10  
        sum += digit ** order  
        temp //= 10  
    if num == sum:  
        print(num)
```

☆ **Write the description below**

The program uses a `for` loop to iterate through each number in the range from 0 to 999. We then convert each number to a string to calculate its length and split it up into its digits, which are cubed and added to a running sum. If the final sum equals the original number, it is an Armstrong number and is printed to the console.

The program uses a for loop to iterate over all numbers in the range of 0 to 999. It then calculates the order of the number using the `len()` function and stores it in the order variable. It initializes the sum variable to 0 and uses a while loop to calculate the sum of the cubes of each digit. Finally, it compares the sum with the original number and prints the number if they are equal.

This program can be easily modified to find Armstrong numbers in a different range by changing the range of the for loop.

The output of this program will be the list of all Armstrong numbers between 0 and 999, which are 0, 1, 153, 370, and 371.